

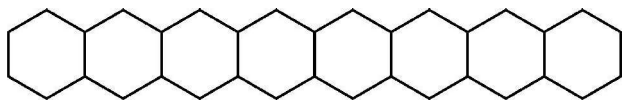


Series : W2YXZ

SET ~ 2



रोल नं.
Roll No.



प्रश्न-पत्र कोड
Q.P. Code **56/2/2**

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

नोट / NOTE



- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।
Please check that this question paper contains 23 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 33 प्रश्न हैं।
Please check that this question paper contains 33 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.



रसायन विज्ञान (सैद्धांतिक) CHEMISTRY (Theory)



निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 70

Maximum Marks : 70

56/2/2

731-2

1

P.T.O.



General Instructions :

Read the following instructions carefully and follow them :

- (i) *This question paper contains 33 questions. All questions are compulsory.*
- (ii) *This question paper is divided into FIVE sections – Section A, B, C, D and E.*
- (iii) *Section A – questions number 1 to 16 are multiple choice type questions. Each question carries 1 mark.*
- (iv) *Section B – questions number 17 to 21 are very short answer type questions. Each question carries 2 marks.*
- (v) *Section C – questions number 22 to 28 are short answer type questions. Each question carries 3 marks.*
- (vi) *Section D – questions number 29 and 30 are case-based questions. Each question carries 4 marks.*
- (vii) *Section E – questions number 31 to 33 are long answer type questions. Each question carries 5 marks.*
- (viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the sections except Section –A.*
- (ix) *Kindly note that there is a separate question paper for Visually Impaired candidates.*
- (x) *Use of calculator is NOT allowed.*

You may use the following values of physical constants wherever necessary :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (m}_e\text{)} = 9.1 \times 10^{-31} \text{ kg.}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg.}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg.}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann's constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$



SECTION – A **$16 \times 1 = 16$**

Question No. 1 to 16 are Multiple Choice type questions carrying 1 mark each.

For questions number 1 to 4, two statements are given – one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below :

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

1. **Assertion (A) :** Vitamin D cannot be stored in our body.

Reason (R) : Vitamin D is fat soluble vitamin and is not excreted from the body in urine.

2. **Assertion (A) :** Aromatic primary amines cannot be prepared by Gabriel Phthalimide synthesis.

Reason (R) : Aryl halides do not undergo nucleophilic substitution reaction with the anion formed by phthalimide.

3. **Assertion (A) :** Cu cannot liberate H_2 on reaction with dilute mineral acids.

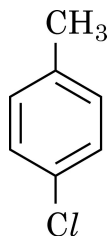
Reason (R) : Cu has positive electrode potential.

4. **Assertion (A) :** In a first order reaction, if the concentration of the reactant is doubled, its half-life is also doubled.

Reason (R) : The half-life of a reaction does not depend upon the initial concentration of the reactant in a first order reaction.



5. Scurvy is caused due to deficiency of
(A) Vitamin B1 (B) Vitamin B2
(C) Ascorbic acid (D) Glutamic acid
6. Nucleotides are joined together by
(A) Glycosidic linkage (B) Peptide linkage
(C) Hydrogen bonding (D) Phosphodiester linkage
7. Which of the following is/are examples of denaturation of protein ?
(A) Coagulation of egg white (B) Curdling of milk
(C) Clotting of blood (D) Both (A) and (B)
8. The conversion of phenol to salicylic acid can be accomplished by
(A) Reimer-Tiemann reaction (B) Friedel-Crafts reaction
(C) Kolbe reaction (D) Coupling reaction
9. What will be formed after oxidation reaction of secondary alcohol with chromic anhydride (CrO_3) ?
(A) Aldehyde (B) Ketone
(C) Carboxylic acid (D) Ester
10. Which is the correct IUPAC name for



- (A) Methylchlorobenzene (B) Toluene
(C) 1-Chloro-4-Methylbenzene (D) 1-Methyl-4-Chlorobenzene

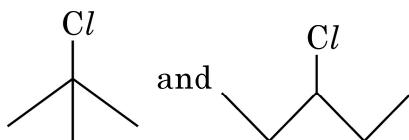


11. The diamagnetic species is :
(A) $[\text{Ni}(\text{CN})_4]^{2-}$ (B) $[\text{NiCl}_4]^{2-}$
(C) $[\text{Fe}(\text{CN})_6]^{3-}$ (D) $[\text{CoF}_6]^{3-}$
[At. No. Co = 27, Fe = 26, Ni = 28]
12. The complex ions $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$ are called
(A) Ionization isomers (B) Linkage isomers
(C) Co-ordination isomers (D) Geometrical isomers
13. The element having $[\text{Ar}]3d^{10}4s^1$ electronic configuration is
(A) Cu (B) Zn
(C) Cr (D) Mn
14. The number of molecules that react with each other in an elementary reaction is a measure of the :
(A) activation energy of the reaction (B) stoichiometry of the reaction
(C) molecularity of the reaction (D) order of the reaction
15. Which among the following is a false statement ?
(A) Rate of zero order reaction is independent of initial concentration of reactant.
(B) Half-life of a zero order reaction is inversely proportional to the rate constant.
(C) Molecularity of a reaction may be zero.
(D) For a first order reaction, $t_{1/2} = 0.693/k$.
16. The charge required for the reduction of 1 mol of MnO_4^- to MnO_2 is
(A) 1 F (B) 3 F
(C) 5 F (D) 6 F



SECTION – B

17. (a) In the following pair of halogen compounds, which compound undergoes S_N1 reaction faster and why ? 1



- (b) Arrange the following compounds in increasing order of their reactivity towards S_N2 displacement : 1
- 2-Bromo-2-methylbutane, 1-Bromopentane, 2-Bromopentane.

18. A reaction is of second order with respect to a reactant. How is the rate of reaction affected if the concentration of the reactant is (i) doubled (ii) reduced to half ? 2

19. When FeCr_2O_4 is fused with Na_2CO_3 in the presence of air it gives a yellow solution of compound (A). Compound (A) on acidification gives compound (B). Compound (B) on reaction with KCl forms an orange coloured (C). An acidified solution of compound (C) oxidises Na_2SO_3 to (D). Identify (A), (B), (C) and (D). 2

20. Explain $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an inner orbital complex whereas $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is an outer orbital complex. [At. No. Co = 27, Ni = 28] 2

21. (A) The rate constant for a zero order reaction $\text{A} \rightarrow \text{P}$ is $0.0030 \text{ mol L}^{-1}\text{s}^{-1}$. How long will it take for the initial concentration of A to fall from 0.10 M to 0.075 M ? 2

OR

- (B) The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if $k = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$? 2

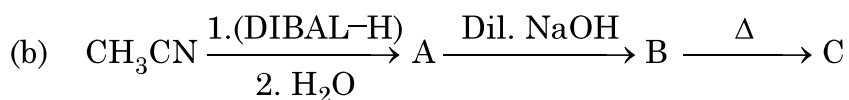
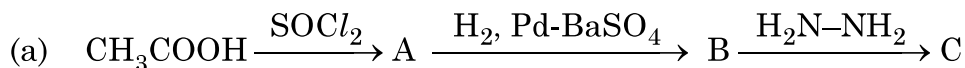


SECTION – C

22. A solution containing 15 g urea (molar mass = 60 g mol⁻¹) per litre of solution in water has the same osmotic pressure (isotonic) as a solution of glucose (molar mass = 180 g mol⁻¹) in water. Calculate the mass of glucose present in one litre of its solution. 3
23. Calculate $\Lambda^\circ m$ for acetic acid and its degree of dissociation (α) if its molar conductivity is 48.1 $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$. 3
Given that
 $\Lambda^\circ m (\text{HCl}) = 426 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 $\Lambda^\circ m (\text{NaCl}) = 126 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 $\Lambda^\circ m (\text{CH}_3\text{COONa}) = 91 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
24. (a) Of the d⁴ species, Cr²⁺ is strongly reducing while Mn³⁺ is strongly oxidising. Why ? 3 × 1
(b) Write two consequences of lanthanoid contraction.
(c) Which element of 3d series has lowest enthalpy of atomisation and why ?
25. Write IUPAC names of the following coordination entities : 3
(a) $[\text{Fe}(\text{en})_2\text{Cl}_2]^+$
(b) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}]\text{SO}_4$
(c) $[\text{Ni}(\text{CN})_4]^{2-}$
26. (A) Explain the following reactions and write chemical equation involved : 3 × 1 = 3
(a) Wolff-Kishner reduction
(b) Etard reaction
(c) Cannizzaro reaction

OR

- (B) Write the structures of A, B and C in the following sequence of reactions : 2 × 1½ = 3

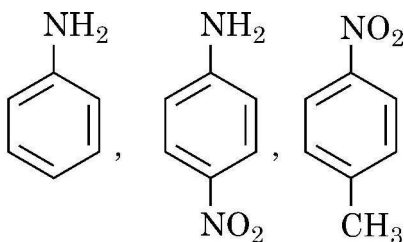




27. (a) Define the following : 2 + 1 = 3
- (i) Enantiomers
- (ii) Racemic mixture
- (b) Why is chlorobenzene resistant to nucleophilic substitution reaction ?
28. (a) Write the product obtained when D-glucose reacts with $\text{H}_2\text{N} - \text{OH}$. 3
- (b) Amino acids show amphoteric behaviour, why ?
- (c) Why vitamin C cannot be stored in our body ?

SECTION – D

29. Amines have a lone pair of electrons on nitrogen atom due to which they behave as Lewis base. Greater the value of K_b or smaller the value of $\text{p}K_b$, stronger is the base. Amines are more basic than alcohols, ethers, esters, etc. The basic character of aliphatic amines should increase with the increase of alkyl substitution. But it does not occur in a regular manner as a secondary aliphatic amine is unexpectedly more basic than a tertiary amine in aqueous solutions. Aromatic amines are weaker bases than ammonia and aliphatic amines. Electron releasing groups such as $-\text{CH}_3$, $-\text{OCH}_3$, $-\text{NH}_2$, etc., increase the basicity while electron-withdrawing substituents such as $-\text{NO}_2$, $-\text{CN}$, halogens etc., decrease the basicity of amines. The effect of these substitute is more at p^- than at m^- position.
- (a) Arrange the following in the increasing order of their basic character. Give reason : 2



- (b) Why $\text{p}K_b$ of aniline is more than that of methylamine ? 1



-
- (c) (i) Arrange the following in the increasing order of their basic character in an aqueous solution : 1



OR

- (c) (ii) Why ammonolysis of alkyl halides is not a good method to prepare pure amines ? 1

30. The spontaneous flow of the solvent through a semipermeable membrane from a pure solvent to a solution or from a dilute solution to a concentrated solution is called osmosis. The phenomenon of osmosis can be demonstrated by taking two eggs of the same size. In an egg, the membrane below the shell and around the egg material is semipermeable. The outer hard shell can be removed by putting the egg in dilute hydrochloric acid. After removing the hard shell, one egg is placed in distilled water and the other in a saturated salt solution. After some time, the egg placed in distilled water swells-up while the egg placed in salt solution shrinks. The external pressure applied to stop the osmosis is termed as osmotic pressure (a colligative property). Reverse osmosis takes place when the applied external pressure becomes larger than the osmotic pressure.

- (a) Define reverse osmosis. Name one SPM which can be used in the process of reverse osmosis. 2
- (b) (i) What do you expect to happen when red blood corpuscles (RBC's) are placed in 0.5% NaCl solution ? 1

OR

- (b) (ii) Which one of the following will have higher osmotic pressure in 1 M KCl or 1 M urea solution. Justify your answer. 1
- (c) Why osmotic pressure is a colligative property ? 1



SECTION – E

31. (A) An organic compound 'A', molecular formula C_2H_6O oxidises with CrO_3 to form a compound 'B'. Compound 'B' on warming with iodine and aqueous solution of NaOH gives a yellow precipitate of compound 'C'. When compound 'A' is heated with conc. H_2SO_4 at 413 K gives a compound 'D', which on reaction with excess HI gives compound 'E'. Identify compounds 'A', 'B', 'C', 'D' and 'E' and write chemical equations involved. 5

OR

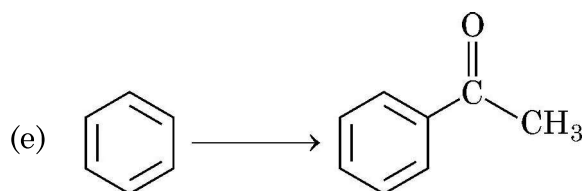
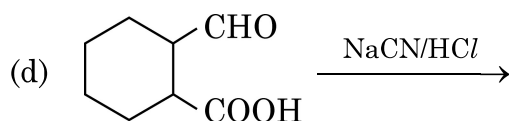
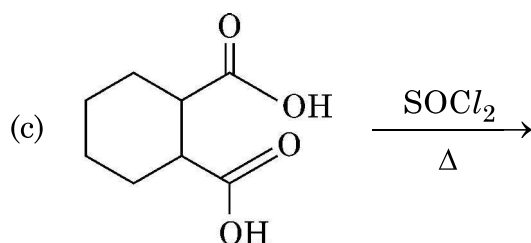
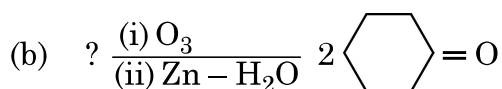
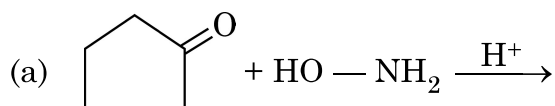
31. (B) (a) Write chemical equations of the following reactions : **3 + 1 + 1 = 5**
- (i) Phenol is treated with conc. HNO_3
 - (ii) Propene is treated with B_2H_6 followed by oxidation by H_2O_2/OH^- .
 - (iii) Sodium t-butoxide is treated with CH_3Cl .
- (b) Give a simple chemical test to distinguish between butan-1-ol and butan-2-ol.
- (c) Arrange the following in increasing order of acid strength :
phenol, ethanol, water
32. (A) (a) Give IUPAC name of $CH_3 - CH = CH - CHO$. **1**
- (b) Give a simple chemical test to distinguish between propanal and propanone. **1**
- (c) How will you convert the following : **3**
- (i) Toluene to benzoic acid
 - (ii) Ethanol to propan-2-ol
 - (iii) Propanal to 2-hydroxy propanoic acid

OR



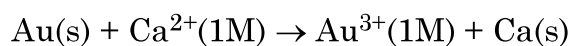
32. (B) Complete each synthesis by giving missing starting material, reagent or products :

5 × 1 = 5



33. (A) (a) Calculate the standard Gibbs energy ($\Delta_r G^\circ$) of the following reaction at 25 °C :

3 + 2



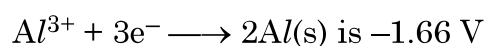
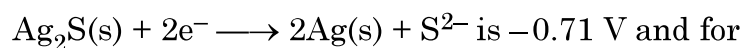
$$E^\circ_{\text{Au}^{3+}/\text{Au}} = +1.5 \text{ V}, E^\circ_{\text{Ca}^{2+}/\text{Ca}} = -2.87 \text{ V}$$

Predict whether the reaction will be spontaneous or not at 25 °C.

$$[1 \text{ F} = 96500 \text{ C mol}^{-1}]$$



- (b) Tarnished silver contains Ag_2S . Can this tarnish be removed by placing tarnished silverware in an aluminium pan containing an inert electrolytic solution such as NaCl ? The standard electrode potential for half reaction :



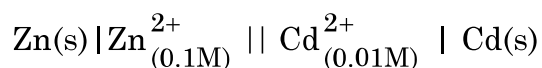
OR

33. (B) (a) Define the following : 2 + 3

(i) Cell potential

(ii) Fuel cell

- (b) Calculate emf of the following cell at 25°C :



Given : $E^\circ_{\text{Cd}^{2+}/\text{Cd}} = -0.40 \text{ V}$

$$E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$$

$$[\log 10 = 1]$$

